[Dry Weight Calculation via InBody]

1. Pre-Analysis of the patient condition

As per the diabetes & hypoalbuminemia condition of dialysis patients, InBody recommends to set three different ECW ratio goals in order to obtain the most optimized dry weight for each patient.

Patient's Condition	ECW Ratio Setting
No Diabetes, No Hypoalbuminemia	**0.385**
Have one (either Diabetes or Hypoalbuminemia)	**0.395**
Have both	**0.405**

**Ref: The optimal ratio of extracellular water to total body water (ECW/TBW) determined by bioelectrical impedance analysis (BIA) for setting dry weight in hemodialysis patients, Sasaki et al, The Journal of Japanese Society for Dialysis Therapy, 4110 : 723 ~730 , 2008 (Japanese Journal)*

2. InBody test after measuring Post-HD weight

After checking the Post-HD weight, try InBody test and print the Body Water Type II result sheet.

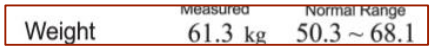
Make sure that InBody test should be done after measuring post-HD weight as the weight will be necessary to be input when starting the InBody Test.



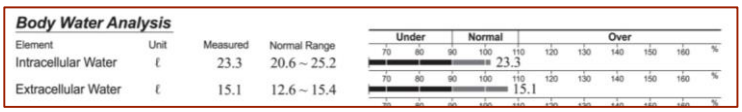
3. Dry Weight calculation with InBody data

You will need

1) **Weight,**
2) **Post-HD ICW & ECW from the InBody result sheet,**
3) **Target ECW Ratio** from Step 1.



	Measured	Normal Range
Weight	61.3 kg	50.3 ~ 68.1



Body Water Analysis

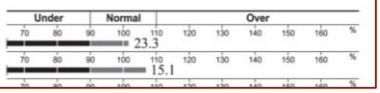
Element	Unit	Measured	Normal Range	Under / Normal / Over
Intracellular Water	ℓ	23.3	20.6 ~ 25.2	23.3
Extracellular Water	ℓ	15.1	12.6 ~ 15.4	15.1

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- Examples

	Measured	Normal Range
Weight	61.3 kg	50.3 ~ 68.1

Body Water Analysis

Element	Unit	Measured	Normal Range
Intracellular Water	ℓ	23.3	20.6 ~ 25.2
Extracellular Water	ℓ	15.1	12.6 ~ 15.4



Example 1) The patient w/o DM, Hypoalbuminemia → Target ECW Ratio: 0.385

ECW ratio means ECW divided by Total Body Water which is sum of ECW and ICW, and in case that the patient has neither Diabetes nor Hypoalbuminemia, the target ECW ratio will be set as **0.385** and it can be expressed as below;

Normal ICW : Normal ECW = 0.615 : 0.385 (* Normal ICW: 1-0.385)

At 'Normal ICW', which means the net ICW without water retention, the post-HD ICW can be input.

23.3(L): Normal ECW = 0.615 : 0.385
Normal ECW = 14.59 (L)

Therefore, normal ECW which the patient is supposed to achieve will be 14.59L. After that, by deducting the obtained normal ECW from the current ECW, the retained ECW will be obtained; 0.51 (L)

(Retained ECW) = (Current post-HD ECW) – (Normal ECW)
= 15.1 – 14.59 = 0.51(L)

Then, InBody-Recommended dry weight will be obtained by subtracting the retained ECW from the post HD weight;

(InBody-recommended Dry Weight) = (Post-HD Weight) – (the Retained ECW)
= 63.1 kg – 0.51 L = 60.79 kg

In conclusion, in accordance with the target ECW Ratio (0.385), the patient will be recommended to achieve 60.79 kg in next time HD, which is the 0.51L less than current weight. In other words, 0.51 L is currently still retained in the patient's body, and this would be advised to remove further.

These calculation flows could be summarized as below;

(Normal ICW) : (Normal ECW) = 0.615 : 0.385
→ 23.3(L) : (Normal ECW) = 0.615 : 0.385
(Normal ECW) = 14.59(L)
(Retained ECW) = (Current ECW)–(Normal ECW) = 0.51(L)

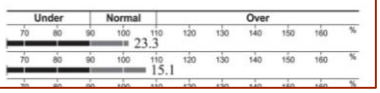
(InBody Recommended Dry Weight)
= (Post-HD Weight) – (the retained ECW) = 60.79 kg

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- Examples

	Measured	Normal Range
Weight	61.3 kg	50.3 ~ 68.1

Body Water Analysis

Element	Unit	Measured	Normal Range
Intracellular Water	ℓ	23.3	20.6 ~ 25.2
Extracellular Water	ℓ	15.1	12.6 ~ 15.4



Example 2) The patient having DM or Hypoalbuminemia → Target ECW Ratio: 0.395

(Normal ICW) : (Normal ECW) = 0.605 : 0.395

→ 23.3(L) : (Normal ECW) = 0.605 : 0.395

(Normal ECW) = 15.21 (L)

(Retained ECW) = (Current ECW)–(Normal ECW) = -0.11(L)

(InBody Recommended Dry Weight)

= (Post HD Weight) – (Retained ECW) = 61.41 kg

Let's suppose that the patient has Diabetes or Hypoalbuminemia. In this case, the target ECW ratio will be set as 0.395, and the recommended dry weight will be calculated as 61.41kg through the same procedure in example 1, which is 0.11 kg higher than the current post-HD weight.

Likewise, in case that the obtained InBody Dry Weight is higher than the current post-HD weight, it can be interpreted that no further water needs to be removed and the water may be over-extracted. Especially, if this type of result is obtained from the patient who is often undergoing hypotension, muscle cramps, or further events during HD, it could be assumed that the body water of the patient might be over-removed so the dry weight may need to be adjusted in accordance with InBody-recommended Dry Weight.

On the other hands, however, there are some patients who don't get any severe symptom and feel no stress despite of having lower post-HD weight than InBody Dry Weight. In that case, it can be inferred that the current HD treatment is successfully done and that the patient's body is currently well handling the water balance internally. Therefore, in that case, the current HD treatment could be maintained in the same prescription unless any further symptom develops.